Equivalence Class Partitioning(ECP)

EXERCISE: Consider Triangle problem : /* Design and develop a program in a language of your choice to solve the triangle problem defined as follows : Accept three integers which are supposed to be the three sides of triangle and determine if the three values represent an equilateral triangle, isosceles triangle, scalene triangle, or they do not form a triangle at all. Derive test cases for your program based on ECP approach, execute the test cases and discuss the results */

Test Case Name :Equivalence class Analysis for triangle problem

Test Data : Enter the 3 Integer Value(a , b And c)

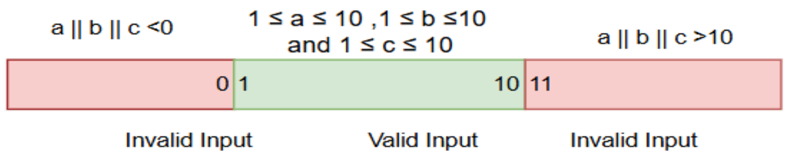
Pre-condition : $1 \leq a \leq 10$, $1 \leq b \leq 10$ and $1 \leq c \leq 10$ and $a < b + c$, $b < a + c$ and $c < a + b$

Brief Description : Check whether given value for a Equilateral, Isosceles , Scalene triangle or can't form a triangle

Source Code:

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a,b,c;

    cout<<"Enter value for side A: ";
    cin>>a;
    cout<<"Enter value for side B: ";
    cin>>b;
    cout<<"Enter value for side C: ";
    cin>>c;

    if(a<=0 || b<=0 || c<=0)
    {
        cout<<"Invalid input: "<<endl;
        return 0;
    }

    if(a>10 || b>10 || c>10)
    {
        cout<<"Invalid Input: "<<endl;
        return 0;
    }

    // Check for a valid triangle using the triangle inequality theorem
    if((a<b+c) && (b<a+c) && (c<a+b))
    {
        if(a==b && b==c)
        {
            cout<<"Equilateral triangle" <<endl;
        }
        else if(a==b || a==c || b==c)
        {
            cout<<"Isosceles triangle"<<endl;
        }
        else
        {
            cout<<"Scalene triangle"<<endl;
        }
    }
    else
    {
        cout<<"Triangle cannot be formed"<<endl;
    }

    return 0;
}
```

EQUIVALENCE CLASS PARTITIONING:



TEST SUMMARY:

Project Name:	Triangle Problem Decision (Equivalence Partitioning)
Test ID:	Solve_Triangle_EP_TES61
Test Title:	Use Equivalence partitioning approach for triangle problem
Test Priority:	Low
Module Name:	Identify_Triangle
Test Data:	Enter the 3 Integer Value (a, b and c)
Designed By:	Urwa Qadir
Designed Date:	2/16/2025
Executed By:	Urwa Qadir
Executed Date:	2/16/2025
Description of Test:	Check whether the given values form an equilateral, isosceles, scalene triangle or no triangle using equivalence
Pre-Conditions:	Range in which testing is to be done: $1 \le a \le 10$, $1 \le b \le 10$ and $1 \le c \le 10$ and $a < b + c$, $b < a + c$ and $c < a + b$
Dependencies:	No dependencies in this test.

TEST CASES:

Test Case	Test Description	Valid			EXpected Output	Actual Output	Status
		A	B	C			
TC_01	Side A is less than 1, which is out of range.	0	5	5	invalid Input	invalid Input	pass
TC_02	Side A is greater than 10, which is out of range.	11	5	5	invalid Input	invalid Input	pass
TC_03	All 3 sides are equal	5	5	5	Equilateral	Equilateral	pass
TC_04	Side B is less than 1, which is out of range.	5	0	5	Invalid Input	Invalid Input	pass
TC_05	Side B is greater than 10, which is out of range.	5	11	5	Invalid Input	Invalid Input	pass

TC_06	Side C is less than 1, which is out of range.	5	5	0	invalid Input	invalid Input	pass
TC_07	Side C is greater than 10, which is out of range.	5	5	11	invalid Input	invalid Input	pass
TC_08	Two sides (A and B) are equal.	5	5	6	Isosceles	Isosceles	pass
TC_09	Two sides (A and C) are equal.	5	6	5	Isosceles	Isosceles	pass
TC_10	Two sides (B and C) are equal.	6	5	5	Isosceles	Isosceles	pass
TC_11	All three sides are different.	10	9	5	scalene	scalene	pass
TC_12	The sum of two sides (B+C=10) is equal to A, violating the triangle inequality theorem.	10	5	5	Not a triangle	Not a triangle	pass
TC_13	The sum of two sides (B+C=8) is less than A, violating the triangle inequality theorem.	10	5	3	Not a triangle	Not a triangle	pass
TC_14	The sum of two sides (A+C=10) is equal to B, violating the triangle inequality theorem.	5	10	5	Not a triangle	Not a triangle	pass
TC_15	The sum of two sides (A+C=8) is less than B, violating the triangle inequality theorem.	5	10	3	Not a triangle	Not a triangle	pass

OUTPUT:

TC-01:

```
Enter value for side A: 0
Enter value for side B: 5
Enter value for side C: 5
Invalid input:
```

TC-2:

```
Enter value for side A: 11
Enter value for side B: 5
Enter value for side C: 5
Invalid Input:
```

TC-03:

```
Enter value for side A: 5
Enter value for side B: 5
Enter value for side C: 5
Equilateral triangle
```

TC-04:

```
Enter value for side A: 5
Enter value for side B: 0
Enter value for side C: 5
Invalid input:
```

TC-05:

```
Enter value for side A: 5
Enter value for side B: 11
Enter value for side C: 5
Invalid Input:
```

TC-06:

```
Enter value for side A: 5
Enter value for side B: 5
Enter value for side C: 0
Invalid input:
```

TC-07:

```
Enter value for side A: 5
Enter value for side B: 5
Enter value for side C: 11
Invalid Input:
```

TC-08:

```
Enter value for side A: 5
Enter value for side B: 5
Enter value for side C: 6
Isosceles triangle
```

TC-09:

```
Enter value for side A: 5
Enter value for side B: 6
Enter value for side C: 5
Isosceles triangle
```

TC-10:

```
Enter value for side A: 6
Enter value for side B: 5
Enter value for side C: 5
Isosceles triangle
```

TC-11:

```
Enter value for side A: 10
Enter value for side B: 9
Enter value for side C: 5
Scalene triangle
```

TC-12:

```
Enter value for side A: 10
Enter value for side B: 5
Enter value for side C: 5
Triangle cannot be formed
```

TC-13:

```
Enter value for side A: 10
Enter value for side B: 5
Enter value for side C: 3
Triangle cannot be formed
```

TC-14:

```
Enter value for side A: 5
Enter value for side B: 10
Enter value for side C: 5
Triangle cannot be formed
```

TC-15:

```
Enter value for side A: 5
Enter value for side B: 10
Enter value for side C: 3
Triangle cannot be formed
```

Lab No. 4b: Write test cases Using Boundary Value Analysis

EXERCISE: recall the triangle problem and write test cases using BVA method using template.

TEST SUMMARY:

Project Name:	Triangle Problem Decision (Boundary Value Analysis)
Test ID:	Solve_Triangle_BVA_TES61
Test Title:	Use Boundary Value Analysis approach for triangle problem
Test Priority:	Low
Module Name:	Identify_Triangle
Test Data:	Enter the 3 Integer Value (a, b and c)
Designed By:	Namra Imtiaz
Designed Date:	2/16/2025
Executed By:	Namra Imtiaz
Executed Date:	2/16/2025

Description of Test:	Check whether the given values form an equilateral, isosceles, scalene triangle or no triangle using equivalence
Pre-Conditions:	Range in which testing is to be done: $1 \le a \le 10$, $1 \le b \le 10$ and $1 \le c \le 10$ and $a < b + c$, $b < a + c$ and $c < a + b$
Dependencies:	No dependencies in this test.

TEST CASES:

TC-ID	**TEST CASE DESCRIPTION**	**INPUT DATA**			**EXPECTED OUTPUT**	**ACTUAL OUTPUT**	**STATUS**
		A	**B**	**C**			
TC-01	A input is not given	X	3	6	Not a triangle	Triangle cannot be formed	pass
TC-02	B input is not given	5	X	4	Not a triangle	Triangle cannot be formed	pass
TC-03	C input is not given	4	7	X	Not a triangle	Triangle cannot be formed	pass
TC-04	C is given in negative	5	5	-1	Not a triangle	Triangle cannot be formed	pass
TC-05	Two sides are same and one side is given different	5	5	1	isosceles	isosceles	pass
TC-06	All sides of inputs are equal	5	5	5	Equilateral	equilateral	pass
TC-07	Two sides are same and one side is given different	5	5	9	isosceles	isosceles	pass
TC-08	The input of C is out of range(i.e range<10)	5	5	10	Not a triangle	Triangle cannot be formed	pass

TC-09	Two sides are same and one side is given different (i.e A and C are 5 and B is 1	5	1	5	Isosceles	isosceles	pass
TC-10	Two sides are same and one side is given different (i.e A and C are 5 and B is 2	5	2	5	Isosceles	isosceles	pass
TC-11	Two sides are same and one side is given different (i.e A and C are 5 and B is 9	5	9	5	Isosceles	isosceles	pass
TC-12	Two sides are same and one side is given different (i.e A and C are 5 and B is out of range	5	10	5	Out of range	Range cannot be formed	pass
TC-13	Two sides are same and one side is given different (i.e B and C are 5 and A is 1	1	5	5	Isosceles	isosceles	pass
TC-14	Two sides are same and one side is given different (i.e B and C are 5 and A is 2	2	5	5	Isosceles	isosceles	pass
TC-15	Two sides are same and one side is given different (i.e B	9	5	5	Isosceles	isosceles	pass

	and C are 5 and A is 9						
TC-16	Two sides are same and one side is given different (i.e B and C are 5 and A is out of range	10	5	5	Not a triangle	Triangle cannot be formed	pass

OUTPUTS:

TC-01

```
Enter Value For A B C : x
Triangle cannot be formed
```

TC-02

```
Enter Value For A B C : 5
x
Triangle cannot be formed
```

TC-03

```
Enter Value For A B C : 4
7
x
Triangle cannot be formed
```

TC-04

```
Enter Value For A B C : 5
5
-1
Triangle cannot be formed
```

TC-05

```
Enter Value For A B C : 5
5
1
Isosceles triangle
```

TC-06

```
Enter Value For A B C : 5
5
5
Equilateral triangle
```

TC-07

```
Enter Value For A B C : 5
9
5
Isosceles triangle
```

TC-08

```
Enter Value For A B C : 5
5
10
Triangle cannot be formed
```

TC-09

```
Enter Value For A B C : 5
1
5
Isosceles triangle
```

TC-10

```
Enter Value For A B C : 5
2
5
Isosceles triangle
```

TC-11

```
Enter Value For A B C : 5
9
5
Isosceles triangle
```

TC-12

```
Enter value for A B C: 5
10
5
Out of range
```

TC-13

```
Enter value for A B C: 1
5
5
Isosceles triangle
```

TC-14

```
Enter value for A B C: 2
5
5
Isosceles triangle
```

TC-15

```
Enter value for A B C: 9
5
5
Isosceles triangle
```

TC-16

```
Enter value for A B C: 10
5
5
Triangle cannot be formed
```